

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)
2025-26, Term-III
Project Problem Statement Submission Form

Sec No- 7A	Team No- 50	Project Title- Patient Registration & EMR Portal
Team Lead Roll Number		2520080063
Team Lead Name		M. Lalitha Madhuhasini
Team Member Roll Number		2520030166
Team Member Name		Simran Jangid

1. Abstract

Healthcare facilities still rely on paper-based or fragmented digital systems for patient registration and medical record management, leading to data loss, delays, and poor accessibility. The Patient Registration & EMR (Electronic Medical Records) Portal is a React-based single-page web application designed to address these challenges. The portal allows patients to register and log in securely, and enables healthcare staff to create, view, and manage electronic medical records efficiently.

The application is built using React.js with functional components, React Router DOM for SPA navigation, React Context API for global authentication state, and React Hook Form with validation pipelines for reliable data entry. The backend is supported by Node.js, Express.js, and MySQL, with Axios handling all API communication. Tailwind CSS is used for responsive and accessible UI design.

The system features two core modules: a Patient Registration & Authentication module handling secure sign-up, login, and form validation; and an EMR Dashboard module providing dynamic routing to patient records, skeleton loading states, and search functionality. The application is optimized for performance using memoization, lazy loading, and code-splitting via Vite, and will be deployed on Netlify/Vercel with Lighthouse-based performance evaluation. This project demonstrates real-world application of front-end engineering principles including component-driven architecture, state management, asynchronous data handling, accessibility compliance, and production-grade deployment workflows.

2. Module Mapping

Module Name	Assigned Roll Number(s)	Signature
Patient Registration & Authentication	2520080063	
Form Engineering & Validation	2520080063	
EMR Dashboard & Record Management	2520030166	
SPA Routing, Protected Routes & Deployment	2520030166	

3. Module Details

M. Lalitha Madhuhasini (2520080063) – Modules: Patient Registration & Authentication, Form Engineering & Validation

Responsible for designing and implementing the Patient Sign-Up and Sign-In pages using React functional components and React Hook Form. Handles all form validation pipelines including field-level error surfaces for inputs such as name, age, contact number, and medical history. Implements React Context API to manage global authentication state across the application. Integrates Axios to communicate with the Node.js/Express backend for registration and login API calls. Ensures controlled form patterns, derived state for submit button activation, and protected route logic based on authentication status.

Simran Jangid (2520030166) – Modules: EMR Dashboard & Record Management, SPA Routing, Protected Routes & Deployment

Responsible for building the EMR Dashboard UI including patient records listing, detail view, and search/filter functionality using React functional components and props-driven architecture. Implements dynamic and nested routing using React Router DOM (e.g., /patients, /patients/:id, /patients/:id/records) along with skeleton UI for loading states and race condition handling for async API calls. Applies memoization and lazy loading for performance optimization. Handles the full build and deployment pipeline using Vite bundler (tree-shaking, minification, asset optimization) and deploys to Netlify/Vercel. Conducts accessibility audits (ARIA, keyboard flow, color contrast) and Lighthouse performance evaluation for production readiness.

4. Frameworks / Platforms / Software Used

Languages: HTML5, CSS3, JavaScript (ES6+), TypeScript

Frontend Framework & Libraries: React.js, React Router DOM, Axios, React Hook Form

Styling: Tailwind CSS

State Management: React Context API

Backend (Supporting Layer): Node.js, Express.js, MySQL

Development Tools: VS Code, Postman, Git & GitHub, npm

Guide Name and Signature (remarks if any) _____